

SYDNEY TECHNICAL HIGH SCHOOL



TRIAL HIGHER SCHOOL CERTIFICATE

2008

Mathematics Extension 2

General Instructions

- Reading time - 5 minutes
- Working time - 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks - 120

- Attempt Questions 1 – 8
- All questions are of equal value

Name : _____

Teacher : _____

Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Question 7	Question 8	Total

Question 1 (15 marks)

a) By completing the square find $\int \frac{dx}{\sqrt{3+2x-x^2}}$ 2

b) Find $\int \sin^2 x \cos^3 x \, dx$ 2

c) Using integration by parts or otherwise find $\int \cos^{-1} x \, dx$ 3

d) Use the substitution $x = 2 \cos \theta$ to find $\int \frac{\sqrt{4-x^2}}{x^2} dx$ 4

e) Use the substitution $t = \tan \frac{\theta}{2}$ to evaluate 4

$$\int_0^{\frac{\pi}{2}} \frac{d\theta}{2 + \cos \theta}$$

Question 2 (15 marks) (Start a new page)

- a) i) Find all pairs of integers x and y such that 3

$$(x + iy)^2 = -3 - 4i$$

- ii) If $x + iy$ with the values of x and y found in part i) 2

are two of the roots of $w^4 = a + ib$, find the two other roots.

- b) Given $z = \sqrt{3} + i$

- i) Find w in the form $x + iy$ if $z + \bar{w} = 2\sqrt{3} - 2i$ 2

- ii) Show that zw is purely imaginary 2

and hence write down the value of $\arg(zw)$

- iii) Write z in modulus – argument form 1

- iv) If the points on the Argand diagram representing z , zw and the origin are the vertices of a triangle find the area of this triangle. 2

- c) i) On an Argand diagram sketch the locus of the point P representing 2

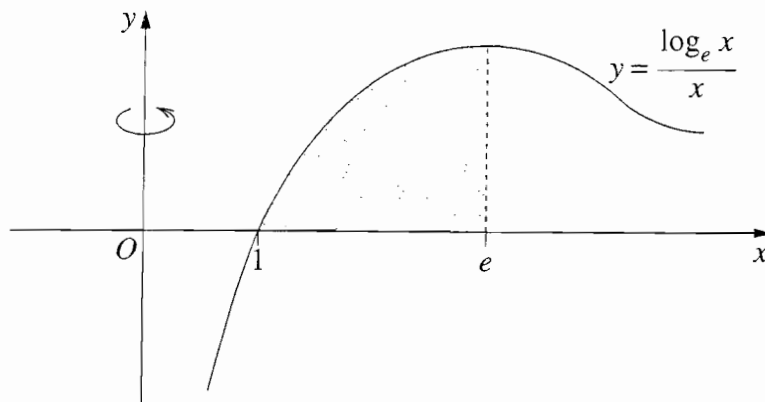
the complex number z such that $|z - (\sqrt{3} + i)| = 1$

- ii) Write down the range of values of $|z|$ if $|z - (\sqrt{3} + i)| = 1$ 1

Question 3 (15 marks) (Start a new page)

- a) The diagram shows the graph of $y = \frac{\log_e x}{x}$.

It has a maximum turning point at $x = e$ and a point of inflection at $x = 2e$.



- i) For each of the following draw a one-third page sketch showing clearly any intercepts and the co-ordinates of any turning points.

$$\alpha) \quad y = \frac{x}{\log_e x} \quad 2$$

$$\beta) \quad y = \frac{d}{dx} \left[\frac{\log_e x}{x} \right] \quad 2$$

$$\delta) \quad y = \frac{\log_e (-x)}{x} \quad 2$$

- ii) Find the coordinates of the maximum turning point on the curve 1

$$y = \log_e \left[\frac{\log_e x}{x} \right]$$

Question 3 is continued on page 5

Question 3 (continued)

- iii) Use the method of cylindrical shells to find the volume of the solid formed 4

when the region bounded by $y = 0$, $y = \frac{\log_e x}{x}$ and $x = e$

is rotated about the y axis.

- b) The curve $x^2 + xy + y^2 = 12$ is symmetrical about $y = x$.

- i) Show that $\frac{dy}{dx} = \frac{-(y+2x)}{x+2y}$ 1

- ii) Find a point (a, b) where $\frac{dy}{dx} = 0$ and explain why there is 3

a vertical tangent at (b, a) .

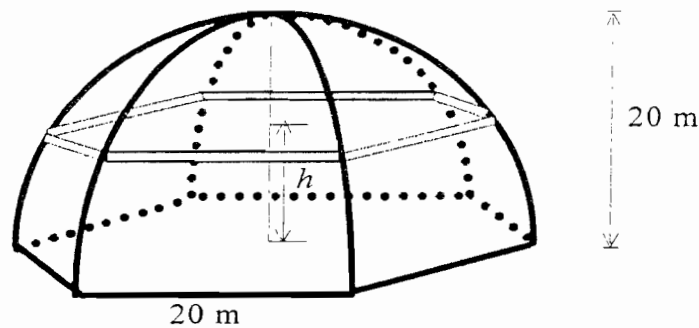
Question 4 (15 marks) (Start a new page)

- a) A dome is sitting on a regular hexagonal base of side 20 metres.

The height of the dome is also 20 metres.

Each strut of the dome (going from the base to the top vertex)

is a quarter of a circle with its centre at the centre of the hexagonal base.



- i) For the slice h metres above the base and parallel to the base, show that 2

the length x of each side of the slice is given by $x = \sqrt{400 - h^2}$ metres.

- ii) Show that the area of the slice described above is given by 2

$$A = \frac{3\sqrt{3}}{2} (400 - h^2) \text{ square metres.}$$

- iii) Hence, or otherwise calculate the volume of the dome. 3

Question 4 is continued on page 7

Question 4 (continued)

b) i) Differentiate $\frac{x^2}{25} + \frac{y^2}{9} = 1$ implicitly 2

ii) Derive the equation of the tangent to the ellipse 2

$\frac{x^2}{25} + \frac{y^2}{9} = 1$ at the point (x_1, y_1) .

iii) Write down the equations of the directrices of $\frac{x^2}{25} + \frac{y^2}{9} = 1$. 1

iv) If $x_1 > 0$ and $y_1 > 0$ find the values of x_1 so that the tangent
at (x_1, y_1) intersects the nearest directrix below the x axis. 3

Question 5 (15 marks) (Start a new page)

a) $(x+1)^2$ is a factor of $P(x) = x^4 + 2x^3 + ax^2 + bx + 4$

i) Find the values of a and b . 3

ii) If $(x - ki)$ is also a factor of $P(x)$ 2
find the value of k **and** the other factor.

b) Let $I_n = \int (\log_e x)^n dx$

i) Show that $I_n = x(\log_e x)^n - n I_{n-1}$ for $n = 1, 2, 3, \dots$ 3

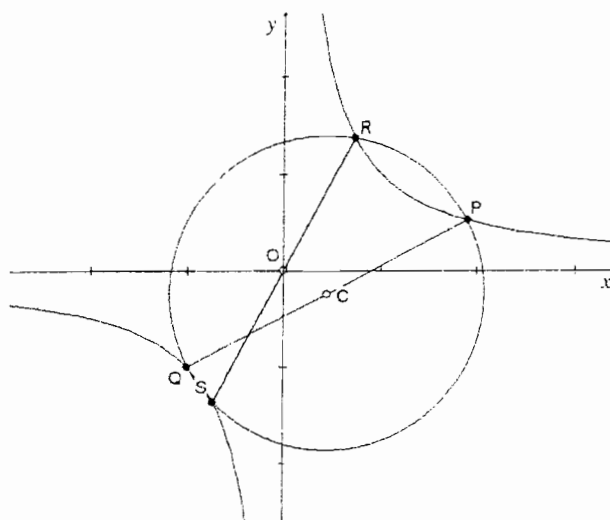
ii) Hence evaluate $\int_1^{e^4} (\log_e x)^2 dx$ 2

Question 5 is continued on page 8

Question 5 (continued)

- c) The circle $(x - h)^2 + (y - k)^2 = r^2$ and the hyperbola $xy = c^2$ intersect at the points P, Q, R and S .

The x coordinates of P, Q, R and S are α, β, γ and δ respectively.



- i) Show that the equation with roots α, β, γ and δ is 2

$$x^4 - 2hx^3 + (h^2 + k^2 - r^2)x^2 - 2c^2kx + c^4 = 0$$

- ii) If the mid point of PQ is the centre of the circle, show that $\alpha + \beta = 2h$ 1

- iii) Hence show that the origin is the mid point of RS . 2

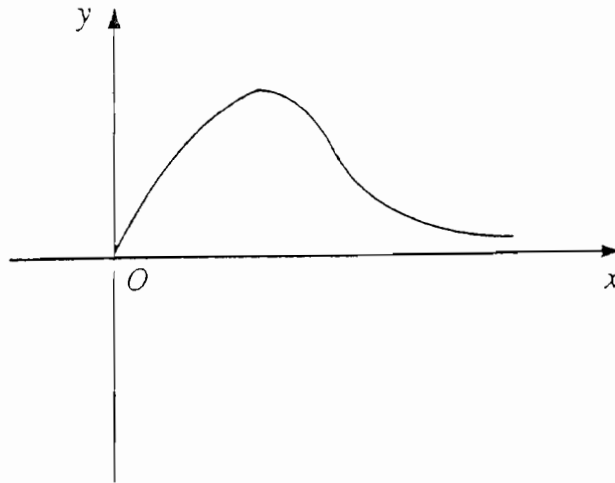
Question 6 (15 marks) (Start a new page)

- a) A particle of unit mass is projected vertically upwards with an initial velocity of 2 m/s . It experiences an air resistance which is numerically equal to gv^2 where g is the acceleration due to gravity.
- i) Explain why the equation of motion is given by $a = -gv^2 - g$ 1
- ii) Find an expression in terms of g 3
for the time taken to reach the maximum height.
- iii) Find an expression in terms of g for the maximum height reached. 3
- b) Given $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$, 3
Use the substitution $x = 4\cos \theta$
to find the exact roots of the equation $x^3 - 12x + 8 = 0$.

Question 6 is continued on page 10

Question 6 (continued)

c)



This is the graph of $f(x) = x^n e^{-x}$ for $x > 0$

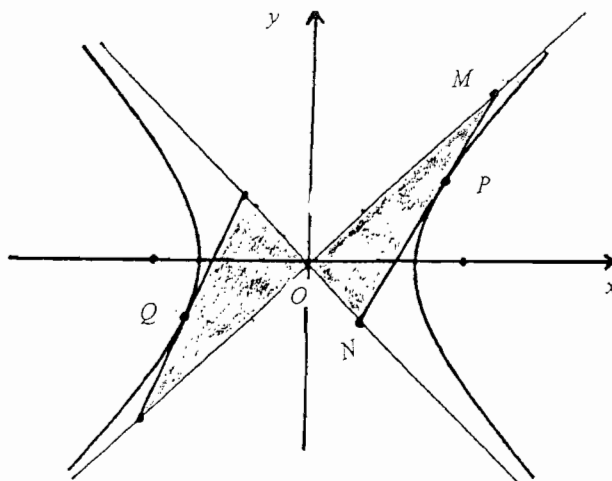
i) Show that the maximum turning point occurs at $x = n$ 2

ii) By considering the values of $f(n)$, $f(n-1)$ and $f(n+1)$ 3

prove that $(1 + \frac{1}{n})^n < e < (1 - \frac{1}{n})^{-n}$

Question 7 (15 marks) (Start a new page)

- a) The diagram shows the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ with tangents drawn at P and Q meeting the hyperbola's asymptotes. O is the origin.



The equation of the tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

at $P(a \sec \theta, b \tan \theta)$ is given by $\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$.

The tangent at $P(a \sec \theta, b \tan \theta)$ meets the asymptotes at M and N as shown in the diagram.

- i) Write down the equations of the asymptotes of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. 1
- ii) Show that the coordinates of M are $\left(\frac{a}{\sec \theta - \tan \theta}, \frac{b}{\sec \theta - \tan \theta} \right)$ 2
- iii) Find the coordinates of N . 1
- iv) Prove that $OM \times ON$ is a constant. 2
- v) Explain why the shaded areas are equal. 1

Question 7 is continued on page 12

Question 7 (continued)

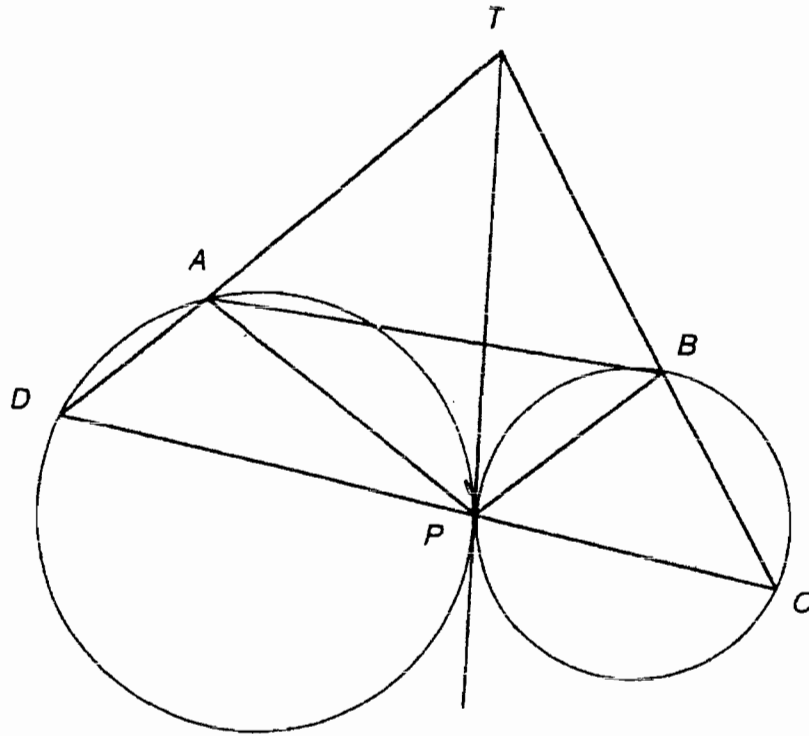
b) i) By solving $\frac{z-1}{z+1} = \cos \theta + i \sin \theta$, show that $z = i \cot \frac{\theta}{2}$ 4

ii) Find all the solutions of $w^6 = -1$ 2

iii) Hence find all the solutions of $(z-1)^6 + (z+1)^6 = 0$ 2

Question 8 (15 marks) (Start a new page)

a)



Circles APD and BPC touch at P .

The point D , P and C are collinear.

TP is the common tangent at P .

TC cuts the circle BPC at B while TD cuts the circle APD at A .

- i) Explain why $\angle TPB = \angle BCP$. 1
- ii) Show that $ATBP$ is a cyclic quadrilateral. 3
- iii) Show that $ABCD$ is a cyclic quadrilateral. 2

Question 8 is continued on page 14

Question 8 (continued)

- b) i) Show that for all values of x and y 1

$$\sin(x + y) - \sin(x - y) = 2 \cos x \sin y$$

- ii) Use mathematical induction to show that for all positive integers n 4

$$\cos x + \cos 2x + \cos 3x + \cdots + \cos nx = \frac{\sin(n + \frac{1}{2})x - \sin \frac{x}{2}}{2 \sin \frac{x}{2}}$$

- iii) Hence show that 4

$$\cos 2x + \cos 4x + \cos 6x + \cdots + \cos 16x = 8 \cos 9x \cos 4x \cos 2x \cos x$$

End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

QUESTION 1

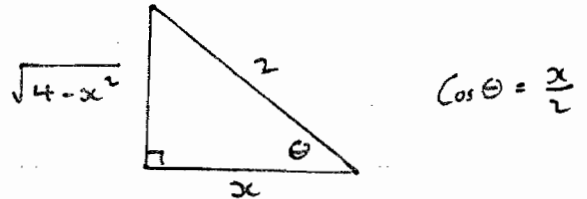
$$\begin{aligned}
 \text{a)} \quad & \int \frac{dx}{\sqrt{3+2x-x^2}} \\
 &= \int \frac{dx}{\sqrt{4-(x-1)^2}} \\
 &= \sin^{-1} \frac{(x-1)}{2} + c
 \end{aligned}$$

$$\begin{aligned}
 \text{b)} \quad & \int \sin^2 x \cos^3 x \, dx \\
 &= \int \sin^2 x (1 - \sin^2 x) \cos x \, dx \\
 &= \int (\sin^2 x - \sin^4 x) \cos x \, dx \\
 &\quad \text{let } u = \sin x \\
 &\quad du = \cos x \, dx \\
 &= \int u^2 - u^4 \, du \\
 &= \frac{1}{3} u^3 - \frac{1}{5} u^5 \\
 &= \frac{1}{3} \sin^3 x - \frac{1}{5} \sin^5 x + c
 \end{aligned}$$

$$\begin{aligned}
 \text{c)} \quad & \int \cos^{-1} x \, dx \\
 &= x \cos^{-1} x - \int x \cdot \frac{-1}{\sqrt{1-x^2}} \, dx \\
 &= x \cos^{-1} x + \int \frac{x}{\sqrt{1-x^2}} \, dx \\
 &= x \cos^{-1} x - \sqrt{1-x^2} + c
 \end{aligned}$$

$$\begin{aligned}
 \text{d)} \quad & \int \frac{\sqrt{4-x^2}}{x^2} \, dx \\
 &\quad x = 2 \cos \theta \\
 &\quad dx = -2 \sin \theta \, d\theta \\
 &= \int \frac{\sqrt{4-4\cos^2 \theta}}{4 \cos^2 \theta} \cdot -2 \sin \theta \, d\theta \\
 &= - \int \frac{\sin^3 \theta}{\cos^2 \theta} \, d\theta
 \end{aligned}$$

$$\begin{aligned}
 &= \int 1 - \sec^2 \theta \, d\theta \\
 &= \theta - \tan \theta \\
 &= \cos^{-1} \frac{x}{2} - \frac{\sqrt{4-x^2}}{x} + c
 \end{aligned}$$



$$\begin{aligned}
 \text{e)} \quad & \int_0^{\frac{\pi}{2}} \frac{d\theta}{2 + \cos \theta} \\
 &\quad t = \tan \frac{\theta}{2} \\
 &\quad d\theta = \frac{2 \, dt}{1+t^2} \\
 &= \int_0^1 \frac{2}{2 + \frac{1-t^2}{1+t^2}} \, dt \\
 &= \int_0^1 \frac{2 \, dt}{2(1+t^2) + 1-t^2} \\
 &= \int_0^1 \frac{2 \, dt}{3+t^2} \\
 &= \frac{2}{\sqrt{3}} \left[\tan^{-1} \frac{t}{\sqrt{3}} \right]_0^1 \\
 &= \frac{2}{\sqrt{3}} \tan^{-1} \frac{1}{\sqrt{3}} \\
 &= \frac{\pi}{3\sqrt{3}}
 \end{aligned}$$

QUESTION 2

$$\begin{aligned}
 \text{a)} \quad \text{i)} \quad & (x+iy)^2 = -3-4i \\
 & x^2 - y^2 + i 2xy = -3-4i
 \end{aligned}$$

$$\begin{aligned}
 \therefore \quad & x^2 - y^2 = -3 \\
 & xy = -2
 \end{aligned}$$

$$\begin{aligned}
 \therefore \quad & x = 1, y = -2 \quad \text{or} \\
 & x = -1, y = 2
 \end{aligned}$$

ii) roots evenly spaced around circle
 $\therefore$ other roots are

$$i(1-2i) \text{ and } i(-1+2i)$$

or

$$2+i \text{ and } -2-i$$

b) i) $\sqrt{3}+i + \bar{w} = 2\sqrt{3}-2i$

$$\bar{w} = \sqrt{3}-3i$$

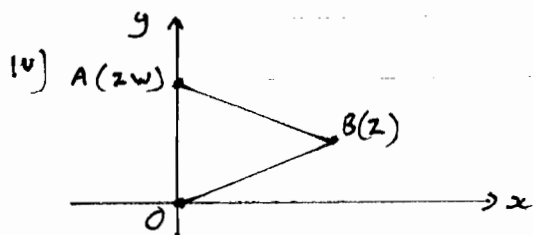
$$\therefore w = \sqrt{3}+3i$$

ii) $zw = (\sqrt{3}+i)(\sqrt{3}+3i)$
 $= 3 + 3\sqrt{3}i + \sqrt{3}i - 3$
 $= 4\sqrt{3}i$

which is purely imaginary

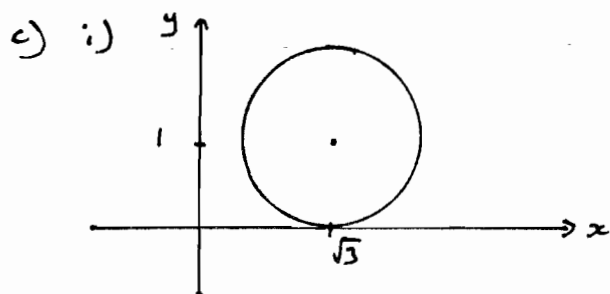
$$\therefore \arg(zw) = \frac{\pi}{2}$$

iii) $z = 2\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)$



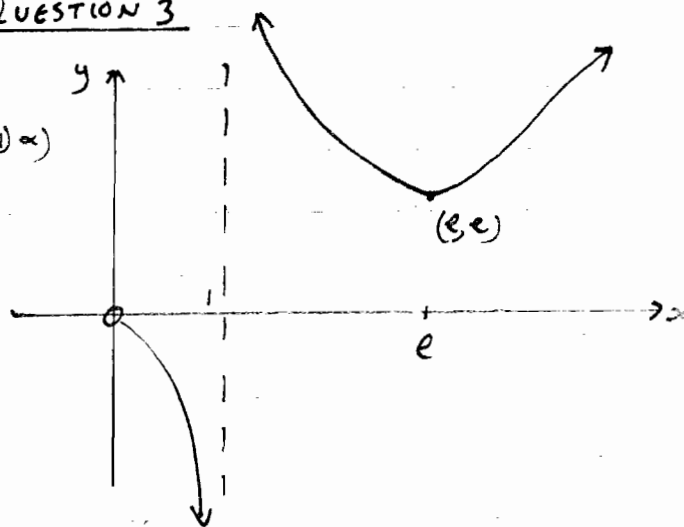
$$\text{Area} = \frac{1}{2} \times 4\sqrt{3} \times \sqrt{3}$$

$$= 6 \text{ sq units}$$

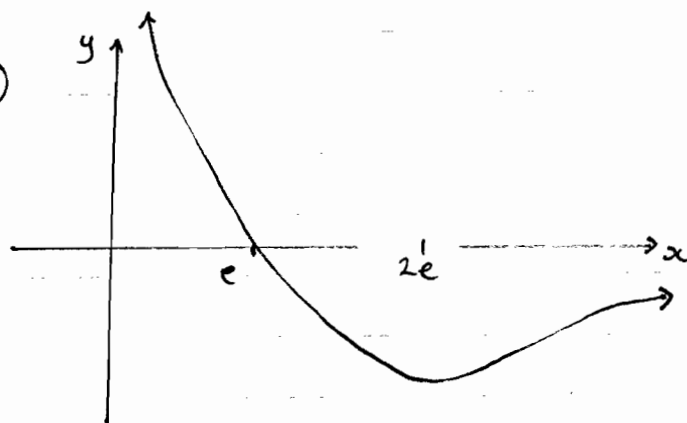


QUESTION 3

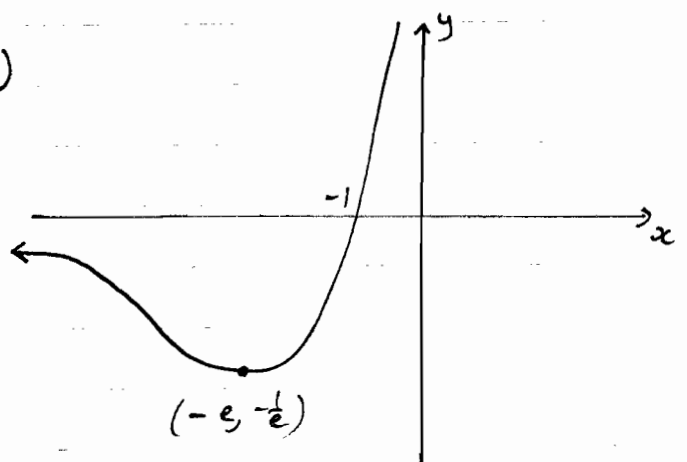
a) i) ∞



β)



δ)



ii) max occurs at $x=e$
 $\therefore y = \log_e \left(\frac{\log_e e}{e} \right)$
 $= -1$

$$\therefore \text{max at } (e, -1)$$

$$\begin{aligned} \text{ii) } \Delta V &= 2\pi x \cdot y \cdot \Delta x \\ &= 2\pi x \frac{\log_e x}{x} \Delta x \end{aligned}$$

$$\begin{aligned} \therefore V &= 2\pi \int_1^e \log_e x \, dx \\ &= 2\pi \left[x \log_e x - x \right]_1^e \\ &= 2\pi \left[(e - e) - (0 - 1) \right] \\ &= 2\pi \text{ cubic units} \end{aligned}$$

$$\text{b) i) } x^2 + xy + y^2 = 12$$

$$2x + x \frac{dy}{dx} + y + 2y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} (x + 2y) = -2x - y$$

$$\frac{dy}{dx} = \frac{-(2x + y)}{x + 2y}$$

$$\text{ii) } \frac{dy}{dx} = 0$$

$$\begin{aligned} \therefore 2x + y &= 0 \\ y &= -2x \end{aligned}$$

$$\therefore x^2 - 2x^2 + 4x^2 = 12$$

$$3x^2 = 12$$

$$x^2 = 4$$

$$x = \pm 2$$

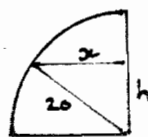
$$\therefore (2, -4) \text{ or } (-2, 4)$$

symmetric about $y = x$ $\therefore$ function is its own inverse

$$\begin{aligned} \therefore f'(b) &= \frac{1}{f'(a)} \quad \left(\frac{dy}{dx} \times \frac{dx}{dy} = 1 \right) \\ &= \frac{1}{1} \end{aligned}$$

QUESTION 4

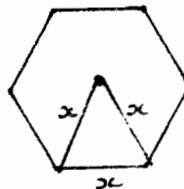
a) i)



$$x^2 + h^2 = 20^2$$

$$\therefore x = \sqrt{400 - h^2}$$

ii)



$$\begin{aligned} A &= 6 \times \left(\frac{1}{2} \times x \times x \times \sin \frac{\pi}{3} \right) \\ &= \frac{3\sqrt{3}x^2}{2} \end{aligned}$$

$$= \frac{3\sqrt{3}}{2} (400 - h^2)$$

$$\begin{aligned} \text{iii) } V &= \frac{3\sqrt{3}}{2} \int_0^{20} (400 - h^2) \, dh \\ &= \frac{3\sqrt{3}}{2} \left[400h - \frac{1}{3}h^3 \right]_0^{20} \\ &= 8000\sqrt{3} \text{ cubic units} \end{aligned}$$

$$\text{b) i) } \frac{2x}{25} + \frac{2y}{9} \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-\frac{2x}{25}}{\frac{2y}{9}}$$

$$\frac{dy}{dx} = \frac{-9x}{25y}$$

$$\text{ii) } m_T = \frac{-9x_1}{25y_1}$$

$$\therefore y - y_1 = \frac{-9x_1}{25y_1} (x - x_1)$$

$$\frac{yy_1}{9} + \frac{xx_1}{25} = \frac{x_1^2}{25} + \frac{y_1^2}{9}$$

$$\frac{yy_1}{9} + \frac{xx_1}{25} = 1$$

$$\text{iii) } b^2 = a^2(1-e^2)$$

$$\therefore e^2 = 1 - \frac{b^2}{a^2}$$

$$= \frac{16}{25}$$

$$e = \frac{4}{5}$$

$$\therefore x = \pm \frac{5}{\frac{4}{5}}$$

$$x = \pm \frac{25}{4}$$

$$\text{iv) Solve simultaneously } \frac{xx_1}{25} + \frac{yy_1}{9} = 1$$

$$\text{and } x = \frac{25}{4}$$

$$\therefore \frac{x_1}{4} + \frac{y_1 y}{9} = 1$$

$$y = \frac{9}{y_1} \left(\frac{4-x_1}{4} \right)$$

we require $y < 0$ and $x_1, y_1 > 0$

$$\frac{9}{y_1} \left(\frac{4-x_1}{4} \right) < 0$$

$$\therefore 4-x_1 < 0$$

$$x_1 > 4$$

$$\therefore 4 < x_1 < 5$$

QUESTION 5

i) $x = -1$ is a double root

$$\therefore p(-1) = 0 \text{ and } p'(-1) = 0$$

$$p(x) = x^4 + 2x^3 + ax^2 + bx + 4$$

$$p'(x) = 4x^3 + 6x^2 + 2ax + b$$

$$p(-1) = 0 \Rightarrow a - b = -3$$

$$p'(-1) = 0 \Rightarrow 2a - b = 2$$

ii) If $x - ki$ is a factor

so is $x + ki$

$\therefore$ roots are $-1, -1, ki, -ki$

using product of roots

$$-(x-1)(x+ki)(x-ki) = 4$$

$$k^2 = 4$$

$$k = \pm 2$$

other factor $(x+2i)$ if $k=2$.

$$\text{b) i) } I_n = \int (\log_e x)^n dx$$

$$u = (\log_e x)^n \quad u' = \frac{n}{x} (\log_e x)^{n-1}$$

$$v = x$$

$$v' = 1$$

$$\therefore I_n = x (\log_e x)^n - \int n (\log_e x)^{n-1} dx$$

$$= x (\log_e x)^n - n I_{n-1}$$

$$\text{ii) } I_2 = \int_1^{e^2} (\log_e x)^2 dx$$

$$= [x (\ln x)^2]_1^{e^2} - 2 I_1$$

$$= 16e^4 - 2 \left[x \ln x \right]_1^{e^2} - I_0$$

$$= 16e^4 - 8e^4 + 2(e^4 - 1)$$

$$= 10e^4 - 2$$

c) i) Solve simultaneously

$$(x-h)^2 + (y-k)^2 = r^2 \quad \text{and}$$

$$y = \frac{c^2}{x}$$

$$(x-h)^2 + \left(\frac{c^2}{x} - k \right)^2 = r^2$$

$$x^2 - 2hx + h^2 + \frac{c^4}{x^2} - \frac{2kc^2}{x} + k^2 = r^2$$

$$x^4 - 2hx^3 + (h^2 + k^2 - r^2)x^2 - 2c^2kx + c^4 = 0$$

ii) mid point is centre of circle

$$\therefore \frac{\alpha + \beta}{2} = h$$

$$\alpha + \beta = 2h$$

iii) sum of roots

$$\alpha + \beta + \gamma + \delta = 2h$$

$$2h + \gamma + \delta = 2h$$

$$\gamma + \delta = 0$$

$$\therefore \frac{\gamma + \delta}{2} = 0$$

$\therefore$ x coordinate of mid point Q.R.U.O.

Q.R passes through the origin

$\therefore$ midpoint is (0,0).

QUESTION 6

a) i) only acceleration acting on particle is gravity and resistance both acting against the motion

$$\therefore a = -g - gv^2$$

$$ii) \frac{dv}{dt} = -g(1+v^2)$$

$$\int \frac{dv}{1+v^2} = \int -g dt$$

$$\tan^{-1} v = -gt + c$$

$$\text{when } t=0 \quad v=2$$

$$\Rightarrow c = \tan^{-1} 2$$

$$\therefore \tan^{-1} v - \tan^{-1} 2 = -gt$$

maximum height when $v=0$

$$\therefore -\tan^{-1} 2 = -gt$$

$$t = \frac{\tan^{-1} 2}{g}$$

$$iii) v \frac{dv}{dx} = -g(1+v^2)$$

$$\int \frac{v dv}{1+v^2} = \int -g dx$$

$$\frac{1}{2} \ln(1+v^2) = -gx + c$$

$$\text{when } x=0 \quad v=2$$

$$\frac{1}{2} \ln 5 = c$$

$$\therefore \frac{1}{2} \ln(1+v^2) - \frac{1}{2} \ln 5 = -gx$$

$$x = -\frac{1}{2g} \ln \left(\frac{1+v^2}{5} \right)$$

$$\text{when } v=0$$

$$x = -\frac{1}{2g} \ln \frac{1}{5}$$

$$b) x^3 - 12x + 8 = 0$$

$$(4 \cos \theta)^3 - 12(4 \cos \theta) + 8 = 0$$

$$64 \cos^3 \theta - 48 \cos \theta + 8 = 0$$

$$8 \cos^3 \theta - 6 \cos \theta + 1 = 0$$

$$4 \cos^3 \theta - 3 \cos \theta = -\frac{1}{2}$$

$$\cos 3\theta = -\frac{1}{2}$$

$$3\theta = \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{8\pi}{3}$$

$$\theta = \frac{2\pi}{9}, \frac{4\pi}{9}, \frac{8\pi}{9}$$

$$\therefore x = 4 \cos \frac{2\pi}{9}, 4 \cos \frac{4\pi}{9}, 4 \cos \frac{8\pi}{9}$$

$$\begin{aligned}
 \text{c) i) } f(x) &= x^n e^{-x} \\
 f'(x) &= nx^{n-1} e^{-x} + x^n (-e^{-x}) \\
 &= e^{-x} (nx^{n-1} - x^n) \\
 &= x^{n-1} e^{-x} (n-x)
 \end{aligned}$$

st pt when $f'(x) = 0$

$$n - x = 0$$

$$x = n$$

$\therefore$ from graph maximum when $x = n$.

ii) from graph

$$f(n-1) < f(n)$$

$$(n-1)^n e^{-(n-1)} < n^n e^{-n}$$

$$\left(\frac{n-1}{n}\right)^n < \frac{e^{-n}}{e^{-(n-1)}}$$

$$\left(\frac{n-1}{n}\right)^n < e^{-1}$$

$$\left(1 - \frac{1}{n}\right)^n > e$$

$$\text{also } f(n+1) < f(n)$$

$$(n+1)^n e^{-n-1} < n^n e^{-n}$$

$$\left(\frac{n+1}{n}\right)^n < \frac{e^{-n}}{e^{-n-1}}$$

$$\left(\frac{n+1}{n}\right)^n < e$$

$$\therefore \left(1 + \frac{1}{n}\right)^n < e < \left(1 - \frac{1}{n}\right)^{-n}$$

QUESTION 7

$$\text{a) i) } y = \pm \frac{b}{a} x$$

$$\text{ii) } y = \frac{b}{a} x$$

$$\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$$

Solve simultaneously

$$\therefore \frac{x \sec \theta}{a} - \frac{bx \tan \theta}{ab} = 1$$

$$x (\sec \theta - \tan \theta) = a$$

$$x = \frac{a}{\sec \theta - \tan \theta}$$

$$\begin{aligned}
 \therefore y &= \frac{b}{a} \cdot \frac{a}{\sec \theta - \tan \theta} \\
 &= \frac{b}{\sec \theta - \tan \theta}
 \end{aligned}$$

$$\therefore M \left(\frac{a}{\sec \theta - \tan \theta}, \frac{b}{\sec \theta - \tan \theta} \right)$$

$$\text{iii) sub } y = -\frac{b}{a} x \text{ into tangent}$$

$$\frac{x \sec \theta}{a} + \frac{bx \tan \theta}{ab} = 1$$

$$x = \frac{a}{\sec \theta + \tan \theta}$$

$$\therefore y = \frac{-b}{\sec \theta + \tan \theta}$$

$$\therefore N \left(\frac{a}{\sec \theta + \tan \theta}, \frac{-b}{\sec \theta + \tan \theta} \right)$$

$$\begin{aligned}
 \therefore OM \times ON &= \frac{\sqrt{a^2+b^2}}{\sec\theta + \tan\theta} \times \frac{\sqrt{a^2+b^2}}{\sec\theta - \tan\theta} \\
 &= \frac{a^2+b^2}{\sec^2\theta - \tan^2\theta} \\
 &= a^2+b^2 \\
 &\text{which is constant}
 \end{aligned}$$

$$v) \text{ Area} = \frac{1}{2} ab \sin C$$

a, b are intercepts OM, ON

$\therefore ab$ is constant

included angle equal - vertically opposite

$\therefore$ areas equal.

$$b) i) \frac{z-1}{z+1} = \cos\theta + i\sin\theta$$

$$z-1 = (z+1)(\cos\theta + i\sin\theta)$$

$$z-1 = z\cos\theta + i z\sin\theta + \cos\theta + i\sin\theta$$

$$z(1 - \cos\theta - i\sin\theta)$$

$$= 1 + \cos\theta + i\sin\theta$$

$$\therefore z = \frac{1 + \cos\theta + i\sin\theta}{1 - \cos\theta - i\sin\theta}$$

$$= \frac{1 + \cos\theta + i\sin\theta}{1 - \cos\theta - i\sin\theta} \times \frac{1 - \cos\theta + i\sin\theta}{1 - \cos\theta + i\sin\theta}$$

$$= \frac{1 - \cos^2\theta + i\sin\theta - i\sin\theta\cos\theta + i\sin\theta + i\sin\theta\cos\theta - \sin^2\theta}{(1 - \cos\theta)^2 + \sin^2\theta}$$

$$(1 - \cos\theta)^2 + \sin^2\theta$$

$$= \frac{1 - (\cos^2\theta + \sin^2\theta) + i 2\sin\theta}{1 - 2\cos\theta + \cos^2\theta + \sin^2\theta}$$

$$= \frac{2i\sin\theta}{2(1 - \cos\theta)}$$

$$= \frac{i 2\sin\frac{\theta}{2} \cos\frac{\theta}{2}}{1 - (1 - 2\sin^2\frac{\theta}{2})}$$

$$= \frac{i \cos\frac{\theta}{2}}{\sin\frac{\theta}{2}}$$

$$= i \cot\frac{\theta}{2}$$

$$ii) w_1 = \cos\frac{\pi}{6} + i\sin\frac{\pi}{6}$$

$$w_2 = i$$

$$w_3 = \cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}$$

$$w_4 = \cos\frac{5\pi}{6} - i\sin\frac{5\pi}{6}$$

$$w_5 = -i$$

$$w_6 = \cos\frac{\pi}{6} - i\sin\frac{\pi}{6}$$

$$iii) (z-1)^6 + (z+1)^6 = 0$$

$$\left(\frac{z-1}{z+1}\right)^6 = -1$$

$$\therefore z = i \cot\frac{\theta}{2}$$

$$\text{where } \theta = \frac{\pi}{6}, \frac{3\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{9\pi}{6}, \frac{11\pi}{6}$$

$$\therefore z = \pm i \cot\frac{\pi}{12},$$

$$\pm i \cot\frac{3\pi}{12},$$

$$\pm i \cot\frac{5\pi}{12} \text{ or}$$

$$z = \pm i \cot\frac{\pi}{12}, \pm i, \pm i \cot\frac{5\pi}{12}$$

Question 8

a) i) $\cancel{TPB} = B\hat{C}P$ (Alternate segment theorem)

ii) Similarly $T\hat{P}A = A\hat{D}P$.

$$A\hat{T}B + A\hat{D}P + B\hat{C}P = 180^\circ \quad (\angle \text{sum } \triangle DCT \text{ is } 180^\circ)$$

$$\therefore A\hat{T}B + T\hat{P}A + T\hat{P}B = 180^\circ$$

$$\therefore A\hat{T}B + A\hat{P}B = 180^\circ \quad (A\hat{P}B \text{ is sum of adjacent } \angle \text{ s } T\hat{P}A, T\hat{P}B)$$

$\therefore ATBP$ is a cyclic quadrilateral (one pair of opposite $\angle$ s supplementary)

(iii)

$$T\hat{B}A = T\hat{P}A \quad (\angle \text{ s at circumference standing on same arc } AT \text{ of circle } ATBP)$$

$$\therefore T\hat{B}A = A\hat{D}P \quad (T\hat{P}A = A\hat{D}P, \text{ proved in (ii)})$$

$\therefore ABCD$ is cyclic (exterior angle equal to interior opposite angle)

b) i)

$$\begin{aligned} \sin(x+y) - \sin(x-y) &= \sin x \cos y + \cos x \sin y - (\sin x \cos y - \cos x \sin y) \\ &= 2 \cos x \sin y \end{aligned}$$

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ii)

If $n = 1$ LHS = $\cos x$

$$\text{RHS} = \frac{\sin\left(\frac{3x}{2}\right) - \sin\left(\frac{x}{2}\right)}{2 \sin\left(\frac{x}{2}\right)}$$

Using i) above RHS = $\frac{2 \cos x \sin\left(\frac{x}{2}\right)}{2 \sin\left(\frac{x}{2}\right)}$

$$= \cos x = \text{LHS}$$

$\therefore$ true for $n = 1$

Assume true for $n = k$

$$\text{i.e. } \cos x + \cos 2x + \cos 3x + \dots + \cos kx = \frac{\sin\left(k + \frac{1}{2}\right)x - \sin\left(\frac{x}{2}\right)}{2 \sin\left(\frac{x}{2}\right)}$$

When $n = k + 1$

$$\cos x + \cos 2x + \cos 3x + \dots + \cos kx + \cos(k+1)x = \frac{\sin\left(k + \frac{1}{2}\right)x - \sin\left(\frac{x}{2}\right)}{2 \sin\left(\frac{x}{2}\right)} + \cos(k+1)x$$

$$= \frac{\sin\left(k + \frac{1}{2}\right)x - \sin\left(\frac{x}{2}\right) + \cos(k+1)x \cdot 2 \sin\left(\frac{x}{2}\right)}{2 \sin\left(\frac{x}{2}\right)}$$

$$= \frac{\sin\left(k + \frac{1}{2}\right)x - \sin\left(\frac{x}{2}\right) + \sin\left((k+1)x + \frac{x}{2}\right) - \sin\left((k+1)x - \frac{x}{2}\right)}{2 \sin\left(\frac{x}{2}\right)} \quad \text{using i) above}$$

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$$= \frac{\sin\left(k + \frac{1}{2}\right)x - \sin\left(\frac{x}{2}\right) + \sin\left(k + \frac{3}{2}\right)x - \sin\left(k + \frac{1}{2}\right)x}{2 \sin\left(\frac{x}{2}\right)}$$

$$= \frac{\sin\left((k+1) + \frac{1}{2}\right)x - \sin\left(\frac{x}{2}\right)}{2 \sin\left(\frac{x}{2}\right)}$$

$\therefore$ True for $n = k + 1$

$\therefore$ Since true for $n = 1$, by induction is true for all positive integral values of $k \geq 1$

iii)

$$\cos 2x + \cos 4x + \cos 6x + \dots + \cos 16x = \cos(2x) + \cos 2(2x) + \cos 3(2x) + \dots + \cos 8(2x)$$

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$$= \frac{\sin\left(8 + \frac{1}{2}\right)2x - \sin\left(\frac{2x}{2}\right)}{2 \sin\left(\frac{2x}{2}\right)}$$

$$= \frac{\sin 17x - \sin x}{2 \sin x}$$

$$= \frac{\sin(9+8)x - \sin(9-8)x}{2 \sin x}$$

$$= \frac{2 \cos 9x \sin 8x}{2 \sin x}$$

Using i) above

$$= \frac{2 \cos 9x \cdot 2 \sin 4x \cos 4x}{2 \sin x}$$

Using double angle on $\sin 8x$

$$= \frac{4 \cos 9x \cdot 2 \sin 2x \cos 2x \cos 4x}{2 \sin x}$$

Using double angle on $\sin 4x$

$$= \frac{8 \cos 9x \cdot 2 \sin x \cos x \cos 2x \cos 4x}{2 \sin x}$$

Using double angle on $\sin 2x$

$$= \frac{8 \cos 9x \cdot \cancel{2 \sin x} \cos x \cos 2x \cos 4x}{\cancel{2 \sin x}}$$

$$= 8 \cos 9x \cos 4x \cos 2x \cos x$$